

Project Design Phase

Solution Architecture

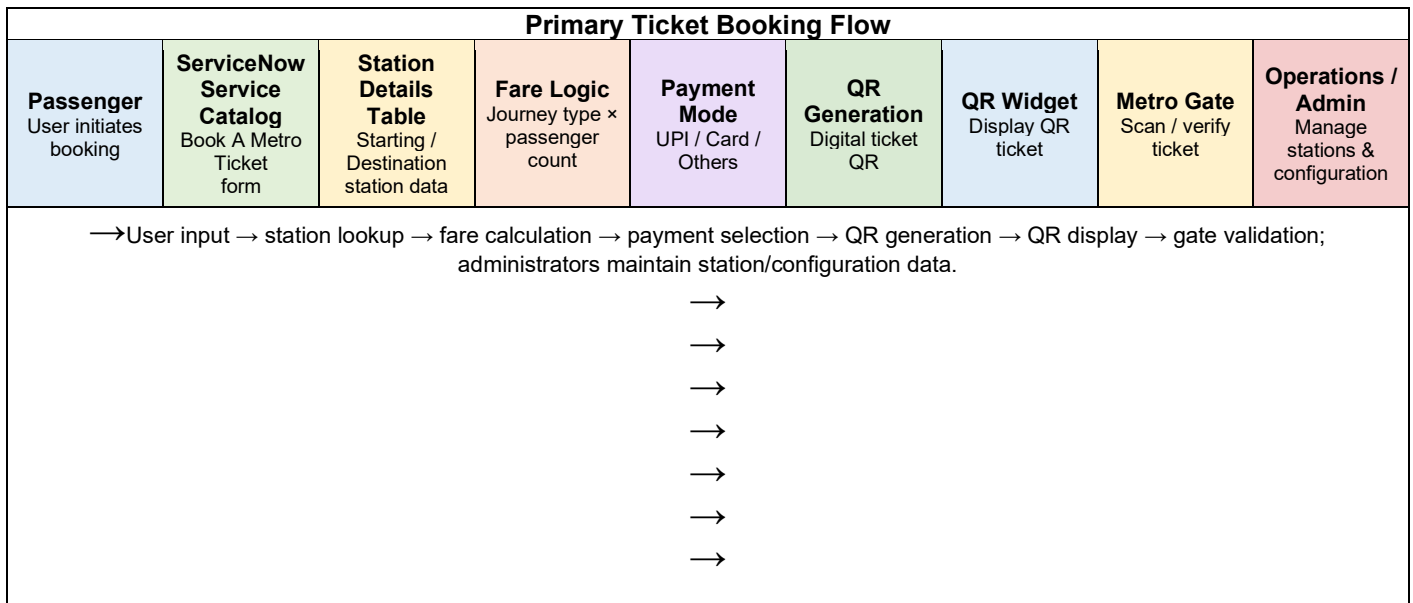
Date	5 September 2026
Team ID	SWUID2026190411
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	4 Marks

Solution Architecture:

The solution architecture of the Metro Ticket Generating System bridges the commuter's ticket-booking requirements with the ServiceNow platform. It defines the main components, data flow, automation and integration used to provide digital metro ticket generation.

- Provide a ServiceNow-based digital solution for metro ticket booking.
- Define the structure and behavior of the ticket-booking system for project stakeholders.
- Automate station selection, journey type, passenger count and fare calculation.
- Generate a QR-based digital ticket and support digital payment selection.
- Provide a design that can be extended with additional stations, routes and integrations.

Architecture and Data Flow:



Main Components and Responsibilities

Component	Technology / ServiceNow Feature	Responsibility
User Interface	ServiceNow Service Catalog	Collects source, destination, journey type, passenger count and payment mode.
Station Data	Custom Metro Station's Details table	Stores metro station information used by the booking form.
Ticket Booking	Book A Metro Ticket catalog item	Provides the passenger-facing ticket booking workflow.

Fare Calculation	Catalog Client Script / fare logic	Calculates the ticket amount based on journey and passenger details.
QR Generation	QR API + ServiceNow client script	Creates the QR image URL for the digital ticket.
QR Display	ServiceNow Portal Widget: Metro QR Widget	Displays the generated QR ticket to the passenger.
Access Control	ServiceNow table ACLs	Controls Create, Read, Write and Delete access for the station table.
Administration	ServiceNow platform	Supports configuration, testing, maintenance and future extensions.

Figure 1: Solution architecture and data flow of the Metro Ticket Generating System in ServiceNow

Reference: Uploaded Solution Architecture template provided for the project design phase.